UNIT IV – CLUSTER ANALYSIS

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1. Definition and Importance of Clustering

Definition

- Clustering is an **unsupervised learning technique**
- It groups similar data objects into clusters

 Goal:

- Maximize **intra-cluster similarity**
- Minimize **inter-cluster similarity**

Example

- Group students based on:
  - Marks
  - Attendance
  - Performance

Importance

- No need for labeled data
- Identifies hidden patterns
- Useful for exploratory analysis

Applications

- Market segmentation
- Image segmentation
- Social network analysis
- Customer behavior analysis

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2. Partitioning Methods

A. K-Means Algorithm

Concept

- Divides data into **K clusters**

Steps

1. Choose K clusters
2. Initialize centroids
3. Assign points to nearest centroid
4. Update centroids
5. Repeat until convergence

Formula (Distance)

```
Euclidean Distance = √((x1-x2)² + (y1-y2)²)
```

Advantages

- Simple and fast
- Works well with large datasets

Limitations

- Sensitive to outliers
- Need to choose K in advance

B. K-Medoids

Concept

- Similar to K-Means but uses **actual data points (medoids)**

Advantages

- Robust to noise and outliers

Difference

Feature	K-Means	K-Medoids
Center	Mean	Medoid
Outliers	Sensitive	Robust

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3. Hierarchical Methods

A. Agglomerative (Bottom-Up)

Steps

1. Each point = separate cluster
2. Merge closest clusters
3. Repeat until one cluster

B. Divisive (Top-Down)

Steps

1. Start with one cluster
2. Split into smaller clusters

Dendrogram

- Tree-like diagram showing cluster hierarchy

Advantages

- No need to specify K
- Easy visualization

Limitations

- High computational cost

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4. Density-Based Methods (DBSCAN)

Concept

- Clusters are formed based on **density of points**

Parameters

- ε (epsilon) → neighborhood radius
- MinPts → minimum points

Types of Points

- Core point
- Border point
- Noise point

Advantages

- Finds arbitrary shaped clusters
- Handles noise well

Limitations

- Sensitive to parameter selection

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5. Grid-Based Methods

A. STING

Concept

- Divides space into rectangular cells

Working

- Statistical information stored in each cell
- Clustering performed using grid

B. CLIQUE

Concept

- Used for **high-dimensional data**

Features

- Combines grid + density approach

Advantages

- Fast processing

- Suitable for large datasets

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6. Cluster Evaluation

A. SSE (Sum of Squared Errors)

Formula

```
SSE = Σ(distance between point and centroid)²
```

□ Lower SSE → better clustering

B. Silhouette Coefficient

Formula

```
S = (b - a) / max(a, b)
```

Where:

- a = intra-cluster distance
- b = nearest cluster distance

Range

- -1 to +1
- Closer to 1 → better clustering

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7. Model-Based Clustering

Concept

- Assumes data generated by **probability distributions**

Example

- Gaussian Mixture Model (GMM)

Advantages

- Flexible
- Can model complex clusters

Limitation

- Computationally expensive

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8. High-Dimensional Data Clustering Challenges

Problems

- Distance becomes less meaningful
- Data sparsity
- Increased complexity

Solutions

- Dimensionality reduction (PCA)
- Feature selection

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9. Graph and Network Clustering

Concept

- Data represented as graph:
  - Nodes → objects
  - Edges → relationships

Example

- Social networks
- Web link analysis

Methods

- Community detection
- Graph partitioning

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10. Constraint-Based Clustering

Concept

- Clustering with user-defined constraints

Types

- Must-link (points must be together)
- Cannot-link (points must not be together)

Example

- Students from same department grouped together

Advantages

- Incorporates domain knowledge

□ Final Summary

- Clustering is key technique for **unsupervised learning**
- Methods include:
  - Partitioning
  - Hierarchical
  - Density-based
  - Grid-based
- Evaluation ensures quality clusters
- Advanced techniques handle complex data